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Bayesian network based general correspondence retrieval method for depth sensing with single-shot structured light☆

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ABSTRACT

This study reinvestigated one of the most fundamental problems in structure light depth sensing field: correspondence retrieval of features between patterns and images. We formulate the global optimum correspondence retrieval by maximizing a conditional probability of correspondence given observed features, which is depicted by a Bayesian network. Different from traditional "code-only" based correspondence retrieval methods, the proposed Bayesian network based method exploits the positional correlations of correspondences of neighboring features, namely, the correspondences of poorly detected features are estimated with the aid of the correspondences of well detected features. The method performs especially well on challenging scenes with rich depth variations, abrupt depth changes, edges, etc. Experiments show that the proposed method increase the correspondence accuracy by about 40% on challenging scenes, compared with traditional "code-only" based correspondence retrieval methods.

1. Introduction

Depth sensing techniques are used widely in many fields, such as robot vision, medical sciences, human–computer interaction, and entertainment. Numerous depth sensing techniques have been developed such as time-of-flight (ToF), stereo vision, and structured light illumination (SLI). Compared with other methods, SLI techniques have advantages in terms of their high accuracy and low cost.

A typical structured light illumination (SLI) depth sensing pipeline involves the following four steps. Firstly, one or multiple coded patterns are delicately designed with special constructions and projected onto a target scene. Then, one or multiple images of the scene with twisted patterns are captured and the features are detected in the images. Subsequently, the correspondences of the features between the images and patterns are retrieved via their predesigned unique codes. Finally, the depth maps are calculated by using triangulation. For SLI techniques, pattern design, feature detection, and correspondence retrieval are the most important problems to be considered.

According to the classification proposed by Salvi [1,2], the SLI techniques can be categorized into two classes: temporal-based and spatial-based schemes. The temporal-based schemes [3–8] encode positional information for a projector's pixels by multiple projected patterns in a time series. This type of method has the advantages in terms of high accuracy and robustness. However, multi-shots must be conducted to capture the pattern series, so the temporal-based techniques are not suitable for dynamic scenes.

The spatial-based schemes only project one pattern and the depth maps are acquired with just a single-shot, which is favorable for depth sensing in dynamic scenes. This type of method uses adjacent pixels to construct unique codes (UC) to distinguish between these pixels. De Bruijn sequence and perfect map are the most commonly used coding maps. A De Bruijn sequence of order m over an alphabet of n symbols is a circular string of length n^m, which contains each substring of length-m exactly once [1]. As a one-dimensional (1D) code, the De Bruijn sequence is usually employed in stripe or slit patterns [9–11], and grid patterns [12–15]. In [16], Vuylsteke constructed a binary chessboard pattern with a 2^6 De Bruijn sequence. The perfect map, which is also called an M-array, perfect sub-map or pseudorandom code [17], is an extension of the De Bruijn sequence in two-dimensional space. Perfect maps are random arrays with $n \times m$ dimensions where a sub-matrix of $r \times v$ dimensions appears only once in the whole pattern [17]. The unique codes in a perfect map are usually constructed by using colors [18–22]

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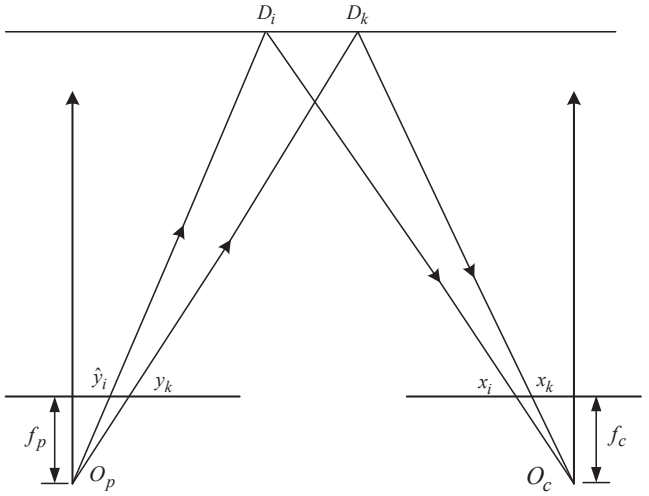


Fig. 1. Estimation of the location of the corresponding feature in a pattern for scenes with fixed depth.

or shapes [23–30]. In the decades, researchers have been concentrating on designing various UC patterns to achieve more accurate and robust feature detection. However, few researches have been made on the vital correspondence retrieval problem. The current dominating method of correspondence retrieval for UC patterns can be regarded as a "code-only" matching (COM) process, which searches in the pattern for the features with the same codes as those in the image as correspondences.

The COM methods may fail to work in the case that features are detected incorrectly or incompletely. Even though a lot of excellent UC patterns have been developed, it is still impossible to perfectly detect features in all scenes. Errors of feature detections occur more frequently in the scenes with rich depth variations, depth discontinuities, edges, or occlusions. This is why depth sensing for these scenes remain one of the most challenging problems in SLI field.

To solve the fore-mentioned challenging problem, we need to improve the traditional COM method instead of optimizing the pattern design. In this study, we propose a general Bayesian network based correspondence retrieval method (BNCR), in order to improve the

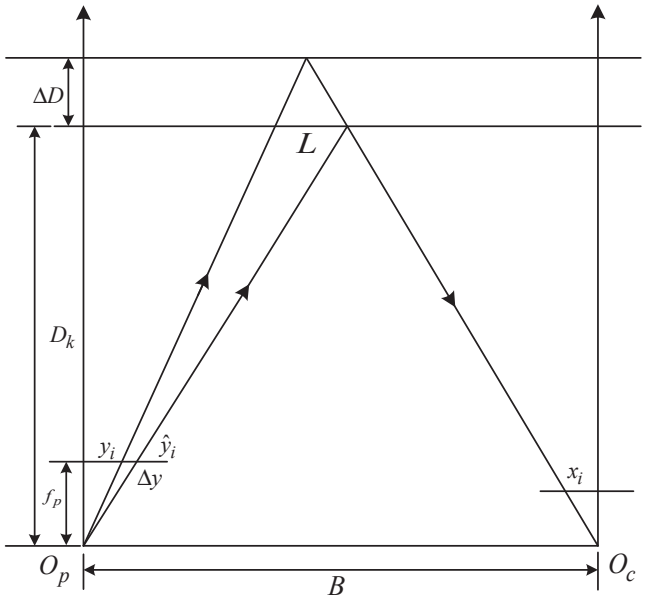


Fig. 2. Estimation of the location of the corresponding feature in a pattern for scenes with non-uniform depth.

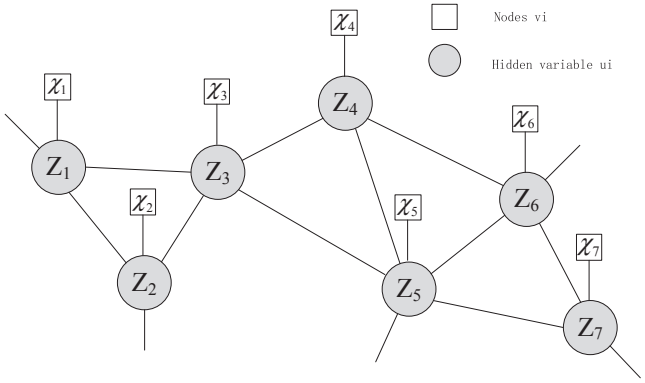


Fig. 3. A Markov network where gray nodes represent hidden variables and white nodes represent observable variables.

correspondence accuracy of the features with low quality (incorrect and incomplete features). The correspondence retrieval process is reinvestigated and formulated as a maximum a posteriori (MAP) estimation. The positional correlation of correspondences between neighboring features is explored and depicted by a Bayesian network. The correspondences of the features with low quality are retrieved by the aid of the correspondences of their neighboring features. The global optimal correspondence, which is the maximum a posteriori probability of the Bayesian network, is determined by the code similarities and the correlations of correspondence between neighboring features. In experiment, we applied our BNCR method to two excellent UC patterns proposed in [16] and [18] to validate its universality. The correspondence results were compared with the traditional COM method, which showed our method greatly improved the correspondence accuracy of the low-quality features in challenging scenes.

2. BNCR

Assuming there are M features in the projected pattern and N features in the captured image, we denote all the M features by $Y = \{Y_1, Y_2 \cdots Y_M\}$ in the projected pattern and all the N features by $X = \{X_1, X_2 \cdots X_N\}$ in the captured image. No matter a UC pattern is constructed by shapes or colors, we could generalize a pattern feature into two components: position and code. Let $Y_j = \left(y_j, \gamma_j\right)$, $j \in \{1, 2 \cdots M\}$ denote a pattern feature, where y_j and γ_j denote its position and encoded code, respectively. It is noted that y_j is scalar, which represents the 1D position along its epipolar line. For an image feature X_i, in addition to position and code, decoding confidence is another important component, which measures how much the decoded code is trustable. Let $X_i = (x_i, \chi_i, \alpha_i)$, $i \in \{1, 2 \cdots N\}$ denote an image feature, where x_i, χ_i and α_i represent the 1D position along its epipolar line, the decoded code

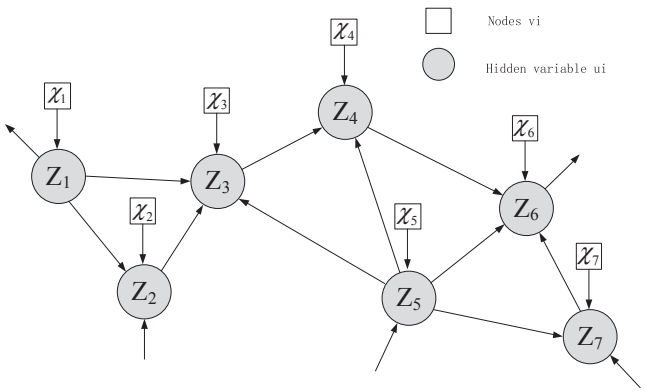


Fig. 4. A Bayesian network.

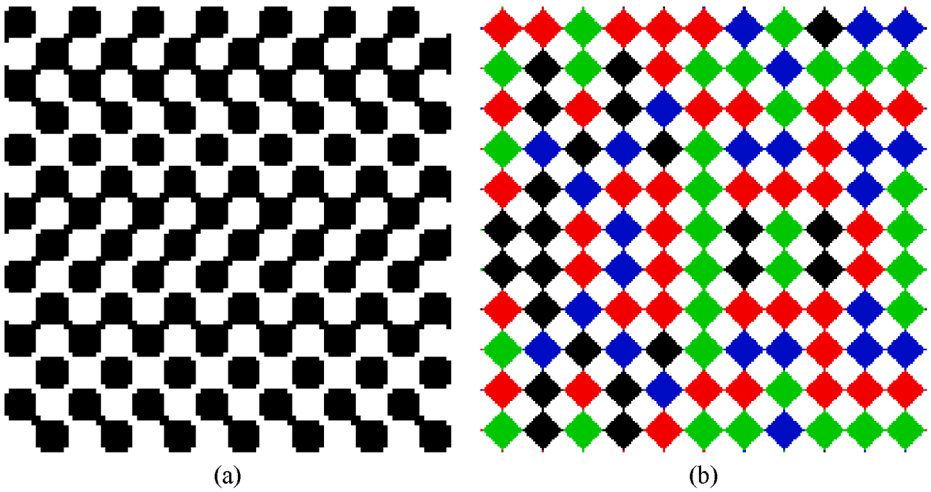


Fig. 5. (a) The chessboard pattern. (b) The rhombus pattern.

and the decoding confidence.

Let $Z_i = f(X_i)$ denote the mapping from an image feature X_i to its corresponding pattern feature Z_i, where $Z_i \in Y, i \in \{1, 2, \cdots, N\}$. If Z_i is regarded as a random variant, retrieving the optimal correspondence is equivalent to maximizing joint conditional probability $P(Z|X)$, where Z denotes all random variants $\{Z_1, \cdots Z_i, \cdots Z_N\}$ and X denotes all observed image features $\{X_1, X_2, \cdots X_N\}$. According to Bayes' theorem, the conditional probability $P(Z|X)$ can be calculated by:

$$P(Z|X) = \frac{P(X|Z)P(Z)}{P(X)}. \tag{1}$$

Considering $P(Z)$ is constant, we have:

$$maxP(Z|X) = maxP(X|Z)P(Z) \tag{2}$$

In Eq. (2), the first term $P(X|Z)$ represents the likelihood of the estimated correspondences and the observation and the second term $P(Z)$ represents joint prior probability of correspondences. The likelihood is measured by the similarities of codes between the observed image features and the estimated corresponding pattern features. Let $d(\chi_i, \gamma_i)$ denote the Hamming distance of code χ_i and γ_i. $P(X_i|Z_i)$ is formulated by:

$$P(X_i|Z_i) = exp(-d(\chi_i, \gamma_i)/\sigma_d), \tag{3}$$

where σ_d is a constant. Obviously, $P(X_i|Z_i)$ is negatively correlated with $d(\chi_i, \gamma_i)$. Given that the Hamming distances $d(\chi_i, \gamma_i)$ are independent between image features, the joint probability $P(X|Z)$ can be calculated by:

$$P(X|Z) = \prod_{i=1}^{N} P(X_i|Z_i) \tag{4}$$

The traditional COM method retrieves correspondence merely by finding pattern features that have the most similar codes with the image features, which mathematically determines all of the Z_i by maximizing Eq. (4). The COM method ignores the prior probability $P(Z)$, which is equivalent to implicitly assuming $P(Z)$ to follow uniform distribution.

In fact, it is unreasonable to assume $P(Z)$ follows uniform distribution. Consider a case that image features X_i and X_k are adjacent and pattern feature Y_k is the correspondence of X_k. Intuitively, the correspondence Z_i of X_i is more likely to be located near Y_k. This comes from the common sense that most parts of a natural scene have smooth depth variations. In other words, there exists correlations of correspondences between neighboring features. Mathematically, let $N(X_i)$ denote the neighboring region of feature X_i. If X_k is located at the neighboring region of X_i, then $X_k \in N(X_i)$. We assume image feature X_k corresponds to pattern feature Y_k, where $Y_k = f(X_k)$, and X_i is located at the neighboring region of X_k, where $X_i \in N(X_k)$. The correspondence of X_k and Y_k determines the depth D_k of a 3D point P_k in the scene.

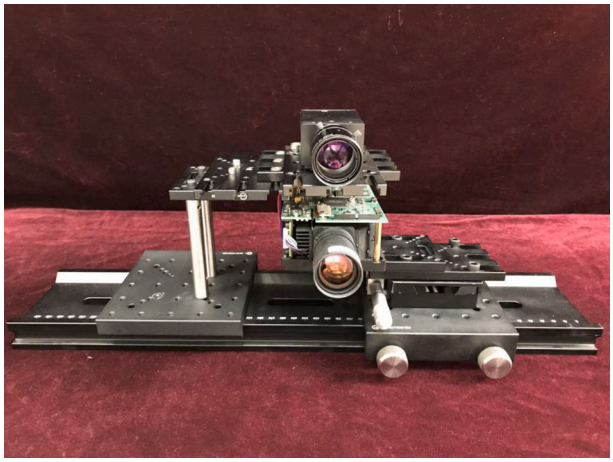


Fig. 6. The experimental platform.

Our following discussion is based on the kind of SLI platform shown in Fig. 1, where the optic axes of both the projector and camera are perpendicular to the baseline. If the region near P_k has constant depth $D_i = D_k$ in the scene, the position $\hat{y}_i$ of the corresponding pattern feature Z_i of X_i can be determined by:

$$\hat{y}_i = f_p/f_c(x_i - x_k) + y_k \tag{5}$$

where f_p and f_c are the focal lengths of the projector and camera, respectively.

Let ΔD denote the depth difference, where $\Delta D = D_i - D_k$. For general case $\Delta D \neq 0$, as shown in Fig. 2, the position y_i of the corresponding pattern feature Z_i varies around $\hat{y}_i$. Let Δy denote the pixel offset of y_i from $\hat{y}_i$. By the similar triangle principle:

$$\frac{\Delta y}{L} = \frac{f_p}{D_k}, \tag{6}$$

$$\frac{L}{B} = \frac{\Delta D}{D_k + \Delta D}, \tag{7}$$

we derive the relationship between Δy and ΔD:

$$\Delta y = f_p B/\left(D_k^2/\Delta D + D_k\right) \tag{8}$$

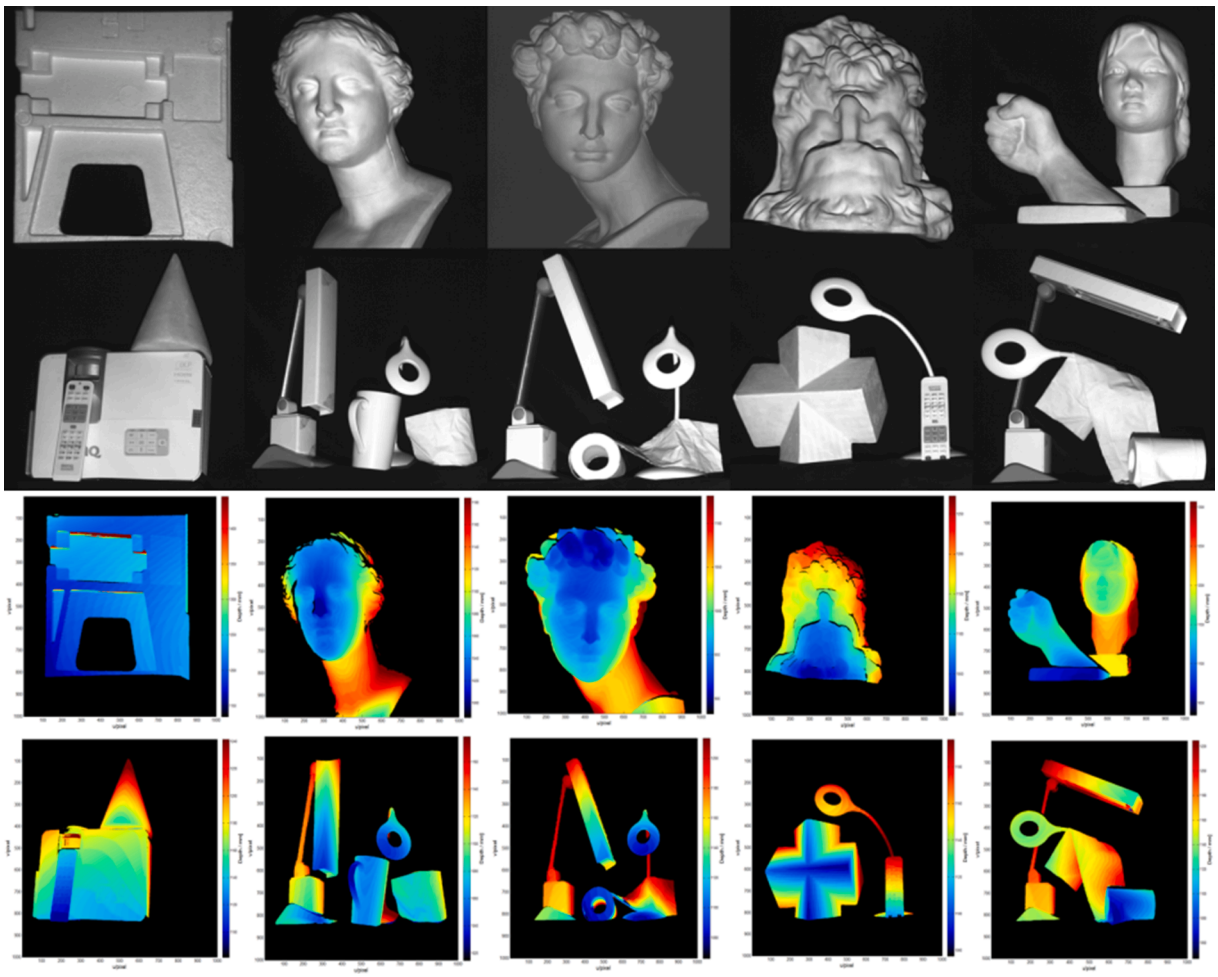


Fig. 7. The test scenes and the ground truth, from top-left to bottom right are scenes from 1 to 10.

Table 1
The error rate (%) comparison of the black-white chessboard pattern [16].

	Scene1	Scene2	Scene3	Scene4	Scene5	Scene6	Scene7	Scene8	Scene9	Scene10
COM in [16]	32.41	17.42	38.03	30.15	41.01	25.31	28.14	38.63	41.01	46.17
Ours	**6.54**	**1.19**	**7.72**	**3.38**	**5.73**	**5.21**	**2.36**	**9.00**	**5.73**	**10.75**

When ΔD is small compared to D_k, Eq. (9) holds:

$$D_k^2/\Delta D \gg D_k. \tag{9}$$

Thus, the offset Δy with respect to $\widehat{y}_i$ is approximately proportional to the depth variation ΔD:

$$\Delta y \approx f_p B \Delta D / D_k^2. \tag{10}$$

We consider a local region in a scene has smooth depth variation, if $|\Delta D| < \Gamma$ holds, where Γ is a threshold. According to Eq. (10), the smooth condition is equivalent to $|\Delta y| < \tau$, where τ is also a threshold. The correlation cost function $g(Z_i, Z_k)$ is defined by:

$$g(Z_i, Z_k) = \begin{cases} 0, |y_i - \widehat{y}_i| \leqslant \tau \\ (|y_i - \widehat{y}_i| - \tau)^2, |y_i - \widehat{y}_i| > \tau \end{cases}, \tag{11}$$

where Z_i and Z_k are the correspondence of X_i and X_k, respectively. The prior probability $P(Z_i|Z_k)$ of Z_i when given correspondence Z_k of neighboring feature X_k is formulated by:

$$P(Z_i|Z_k) = exp\left(-g(Z_i, Z_k)/\sigma_g\right), \tag{12}$$

where σ_g is a constant. Thus, the prior probability $P(Z_i)$ is obtained by comprising the correspondences of all the neighboring features:

$$P(Z_i) = \prod_{k \in N(i)} exp\left(-g(Z_i, Z_k)/\sigma_g\right) \tag{13}$$

The MAP of Eq. (2) can be written as Eq. (14) or (15):

$$maxP(Z|X) = max \prod_{i=1}^{N} exp(-d(\chi_i, \gamma_i)/\sigma_d) \prod_{i=1}^{N} \prod_{k \in N(i)} exp\left(-g(Z_i, Z_k)/\sigma_g\right), \tag{14}$$

$$\max \log P(Z|X) = min \sum_{i=1}^{N} d(\chi_i, \gamma_i) + \sum_{i=1}^{N} \sum_{k \in N(i)} \lambda g(Z_i, Z_k), \tag{15}$$

where λ is a tuning parameter.

One effective way to solve Eq. (14) or (15) is constructing a Markov network and calculating the MAP of the Markov network via loopy belief propagation (BP). The Markov network is shown in Fig. 3. The gray

Table 2
The error rate (%) comparison of the color rhombus pattern [18].

	Scene1	Scene2	Scene3	Scene4	Scene5	Scene6	Scene7	Scene8	Scene9	Scene10
COM in [18]	27.72	14.32	25.4	32.65	39.28	21.81	27.00	35.1	39.28	51.49
Ours	**6.23**	**2.52**	**4.85**	**4.86**	**6.87**	**4.11**	**4.07**	**9.71**	**6.87**	**12.07**



Fig. 8. The visualization of the wrong matched points with chessboard template. The top two rows are the result of COM based technique [16], and the bottom two rows are the results of the proposed BNSR technique, the wrong matched points are marked by yellow circle.

vertices v_i are hidden variables representing the correspondences Z_i of image features X_i. White vertices u_i are observable variables representing the observed codes χ_i of X_i. Each u_i is only connected with one v_i by an edge (v_i, u_i). The similarities of codes χ_i with codes γ_i of Z_i are measured by Hamming distance $d(\chi_i, \gamma_i)$. Edge (v_i, v_k) connecting vertices v_i and v_k represents the correlation between neighboring image features X_i and X_k, where $X_k \in N(X_i)$. The correlation is measured by $g(Z_i, Z_k)$ as given in Eq. (11). For details of loopy BP, please refer to literatures [31,32].

In [32], Sun modeled the problem of stereo matching as a Markov network and solved the optimum by loopy BP. Satisfactory results can always be achieved by using loopy BP, however, the method is computation expensive. Markov network may be not the most appropriate model for correspondence retrieval in SLI. Different from stereo vision, SLI methods could project a predesigned pattern, thus it is easy to build a criterion to determine decoding confidences α_i and evaluate the qualities of the detected features. For features with large α_i, their correspondences are trusted via simple code matching. While for features with small α_i, we need to retrieve their correspondences by code matching and the aid of their neighboring features. Mathematically, we define $v_j \in V(v_i)$, if $v_j \in N(v_i)$ and $\alpha_j > \alpha_i$. Each Z_i can be determined by Eq. (16):

$$Z_i = \underset{Z_i}{argmax} P(Z_i|X_i) = \underset{Z_i}{argmax} P(X_i|Z_i) \prod_{j \in V(i)} P(Z_i|Z_j) P(Z_j), \tag{16}$$

where $P(X_i|Z_i)$ and $P(Z_i|Z_j)$ are formulated by Eq. (3) and (12). Obviously, for a v_i where $V(v_i) = \varnothing$, the correspondence Z_i is calculated by:

$$Z_i = \underset{Z_i}{argmax} P(X_i|Z_i) \tag{17}$$

In fact, solving the optimum correspondences Z_i for all image features is equivalent to calculating the MAP of a Bayesian network. As shown in Fig. 4, the direction of edge (v_i, v_j) is from v_i to v_j, if $\alpha_i > \alpha_j$, and the direction of edge (u_i, v_i) is from u_i to v_i. Obviously, there is no circle in the graph. It is simple and efficient to solve the MAP of Bayesian network. We could

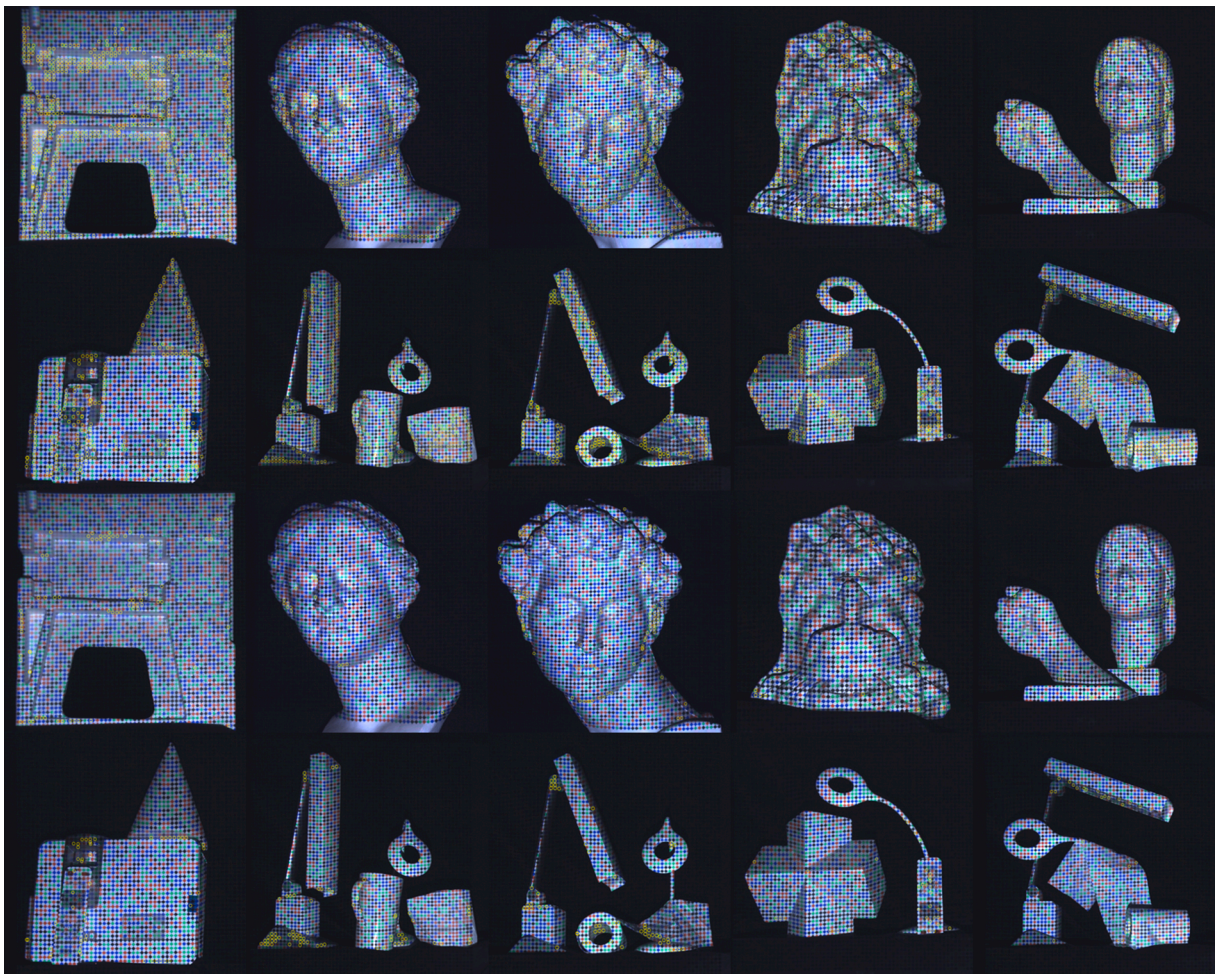


Fig. 9. The visualization of the wrong matched points with chessboard template. The top two rows are the result of COM based technique[16], and the bottom two rows are the results of the proposed BNSR technique, the wrong matched points are marked by yellow circle.

sort all of the image features according to their values of$\alpha_i$and traverse each feature to calculate the optimum correspondences via Eq. (15).

3. Experiments

3.1. System setup

Experiments were conducted to verify the effectiveness and universality of the proposed BNCR method. For universality, we applied BNCR to two excellent and representative based methods proposed by Vuylsteke [16] and Song [18] respectively, and compare the performances with the original COM based correspondence retrieval methods in [16] and [18].

The template used in [16] and [18] are shown in Fig. 5(a), the pattern proposed in [16] is a chessboard-like one, where every grid point is marked with a bright or a dark spot representing the binary states 1 and 0. Every 2×3 window of spots forms a unique 6-bit De Bruijn code. As the author claimed, the method is quite insensitive to reflectance variation, geometric distortion due to oblique surface inclination, and blur of the projected pattern. The pattern proposed in [18] is constructed by chromatic rhombuses with four colors: red, green, blue, and black, as shown in Fig. 5(b). Similarly, every2×3window of rhombuses represents a unique 6-bit pseudorandom code. As Song claimed, the use of rhombic elements has the advantage that adjacent pattern elements do not share a boundary, and thus color confusion in the image between adjacent pattern elements is avoided. The two patterns were projected onto several scenes, and then the captured features were detected via the methods proposed in the both literatures.

The method of Gray code [3] was used to acquire the ground truth. We define the correspondence accuracy by the ratio of the number of features with correct correspondences over the total number of detected features in a captured image. A correspondence is regarded as "correct", if the position of the corresponding feature is less than 3 pixels away from the ground truth.

The experimental platform is shown in Fig. 6, which comprised a digital light processing (DLP) projector with 912×1140 resolution (Texas Instruments DLPLCR4500) and a camera with 2048×2048 resolution (Point Grey Grasshopper3). The optical axes of the camera and the projector were adjusted so they were approximately parallel. The intrinsic and extrinsic parameters of the camera and the projector were calibrated in advance using Zhang's method [33–36]. The baseline was 100 mm.

3.2. Test scenes

We set up ten different scenes which include planes, sculptures and real objects like paper, lamp, projector, etc. The ten scenes and the ground truth are illustrated in Fig. 7.

All the scenes contain rich depth variation, depth discontinuities, edges and occlusions. Only white objects were used in the scenes in order to avoid the disturbances of colors to the feature detection of the color-encoding method in [18]. An investigation of the robustness of

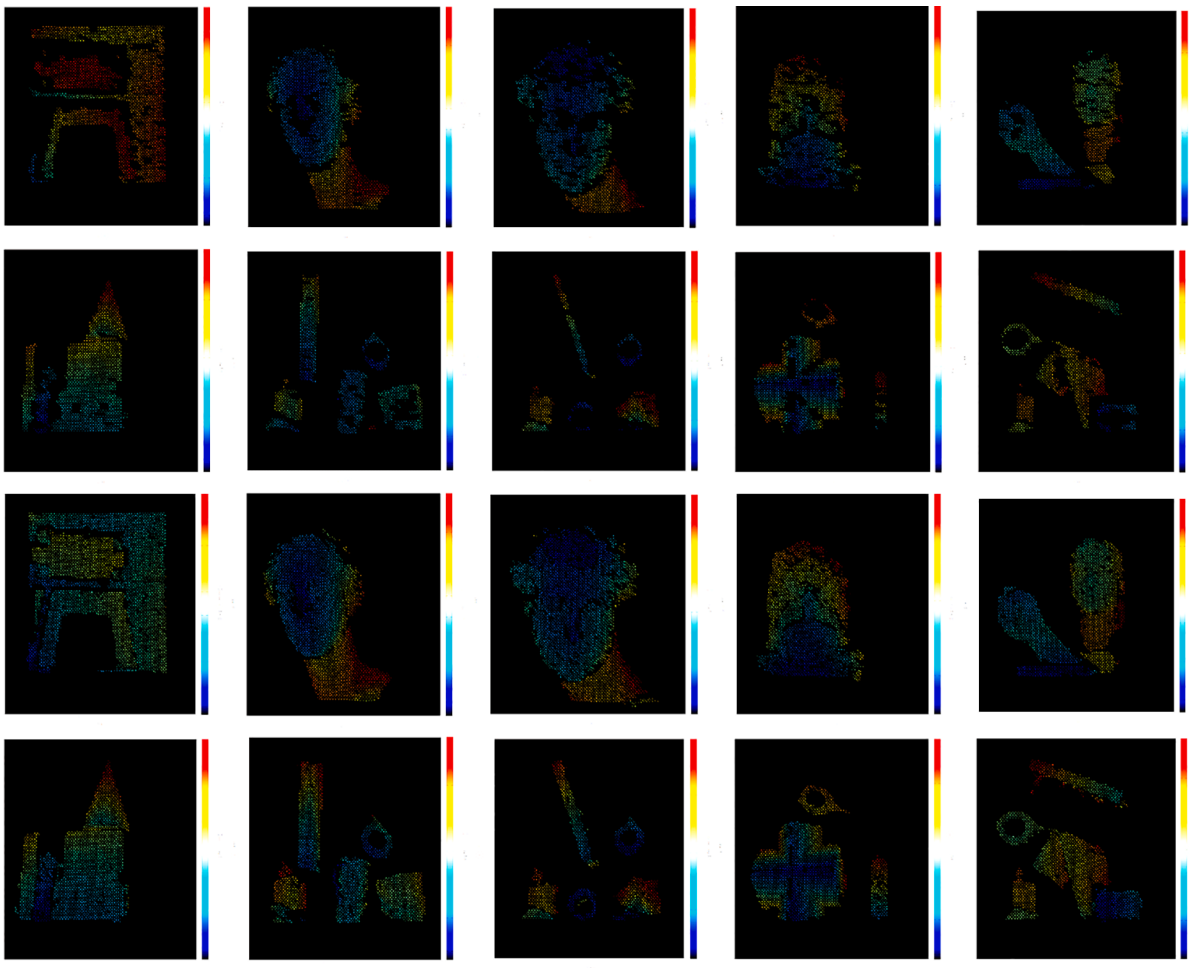


Fig. 10. The final depth map of using the chessboard template. The top two rows are the result of COM based technique [16], and the bottom two rows are the results of the proposed BNSR technique.

color-encoding methods to colorful scenes is beyond the scope of this study.

3.3. Comparison

In Tables 1 and 2 we list the matching error rate of different scenes using chessboard and rhombus pattern, respectively. It is not surprised to observe that using the color rhombus pattern [18] can slightly improve the accuracy of depth sensing of the chessboard pattern. However, with the proposed BNCR technique, the error rate of the chessboard pattern decreases significantly, even achieves much better performances than the color pattern. In addition, as illustrated in Table 2, the BNCR technique also significantly improves the matching performances of the color rhombus pattern. This may reveal a fact that instead of designing more sophisticated codes, it is more efficient to investigate the error resilience technique for the SLI based depth sensing

In Figs. 8 and 9, we visualize the wrong matched points of chessboard and rhombus pattern respectively and the errors are marked by yellow circle in the figures. It can be observed that wrong matched points always appear in the boundary of the objects for the COM based techniques. This is because in the boundary region, the depth is highly discontinues and the emitted codes are prone to missing or error. Taking advantage of the consideration of spatial correlation, the BNCR technique can fix most of the errors. However, the BNCR error still appears at some narrow boundaries like the bracket of the lamp. This is because at these regions, almost all the codes contain errors and it is intractable to find a perfect prior to fix these points.

Finally, in Figs. 10 and 11, we visualize the final depth map of chessboard and rhombus pattern respectively. It can be observed that due to the large error-rate, the depth map of the two COM based techniques appears sparser than the BNCR depth map and object boundary is irregular. Taking advantage of the Bayesian network, the proposed BNCR technique can retrieve a more dense depth map with sharp boundaries and smooth background.

4. Conclusion

We have presented a robust general correspondence retrieval method based on Bayesian network (BNCR) to improve the correspondence accuracies of captured and projected features. The correlations of correspondences of neighboring features are investigated and depicted by a Bayesian network. The optimum correspondences are retrieved by

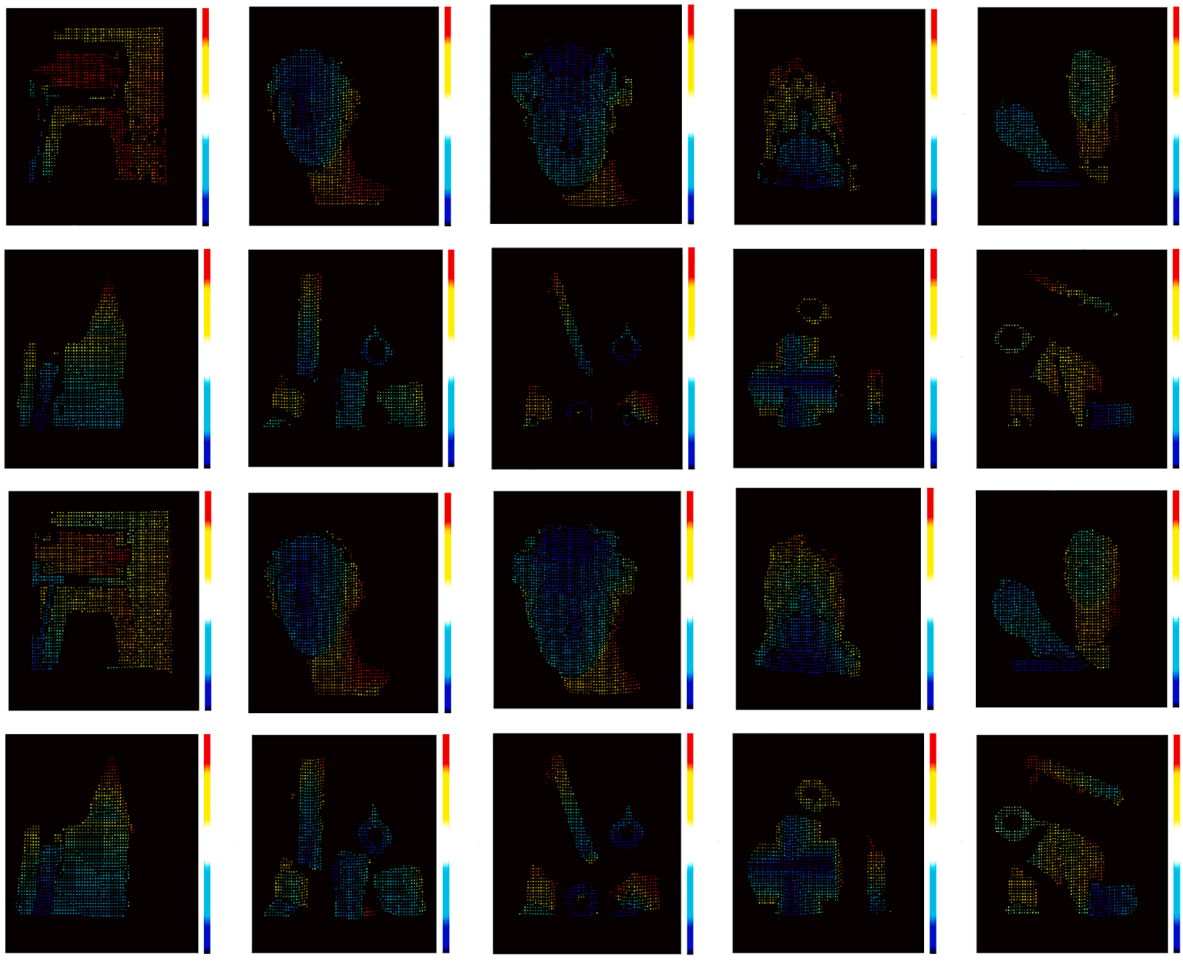


Fig. 11. The final depth map of using the chessboard template. The top two rows are the result of COM based technique [18], and the bottom two rows are the results of the proposed BNSR technique.

maximizing the MAP of the Bayesian network. The proposed method greatly improves the correspondence accuracies of features with low qualities, thus performs well on challenging scenes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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